

Part C

Course title: SC7 Bayes Methods
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1. Let $\pi(\theta)$, $\theta \in \mathbb{R}^p$ be a prior for a parameter θ and let $p(y|\theta)$, $y \in \mathbb{R}^n$ be an observation model. Let $S : \mathbb{R}^n \rightarrow \mathbb{R}^p$ be a p -dimensional summary statistic for data y . Denote by $y_{obs} \in \mathbb{R}^n$ the observed data and let $s_{obs} = S(y_{obs})$. For $s \in \mathbb{R}^p$ let $d(s, s_{obs})$ be a distance measure over $\mathbb{R}^p$ and for $\delta > 0$ let $\Delta_\delta(y_{obs}) = \{y \in \mathbb{R}^n : d(S(y), s_{obs}) < \delta\}$. Let $p(\Delta_\delta(y_{obs})|\theta) = \int_{\Delta_\delta(y_{obs})} p(y|\theta) dy$ and let $p(\Delta_\delta(y_{obs})) = \int_{\mathbb{R}^p} p(\Delta_\delta(y_{obs})|\theta) \pi(\theta) d\theta$.

(a) [6 marks] Suppose $\theta \sim \pi(\cdot)$ and $Y \sim p(\cdot|\theta)$.

- (i) Write down the ABC posterior $\pi_{ABC}(\theta|y_{obs}) = \pi(\theta|Y \in \Delta_\delta(y_{obs}))$ in terms of the quantities defined above.

Solution: (B/2)

$$\pi_{ABC}(\theta|y_{obs}) = \frac{p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)}{p(\Delta_\delta(y_{obs}))}.$$

(ii) Show that

$$\pi_{ABC}(\theta|y_{obs}) = \int_{\Delta_\delta(y_{obs})} \pi(\theta|y)p(y|Y \in \Delta_\delta(y_{obs})) dy,$$

where $p(y|Y \in \Delta_\delta(y_{obs})) = p(y)\mathbb{I}_{y \in \Delta_\delta(y_{obs})}/p(\Delta_\delta(y_{obs}))$ and $p(y) = \int_{\mathbb{R}^p} p(y|\theta)\pi(\theta)d\theta$.

Solution: (B/4) Substituting $p(y|Y \in \Delta_\delta(y_{obs}))$ into the integral on the RHS,

$$\begin{aligned} \int_{\Delta_\delta(y_{obs})} \pi(\theta|y)p(y|Y \in \Delta_\delta(y_{obs})) dy &= \int_{\Delta_\delta(y_{obs})} \frac{\pi(\theta|y)p(y)}{p(\Delta_\delta(y_{obs}))} dy \\ &= \frac{\int_{\Delta_\delta(y_{obs})} p(y|\theta)\pi(\theta) dy}{p(\Delta_\delta(y_{obs}))} \\ &= \frac{p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)}{p(\Delta_\delta(y_{obs}))} \\ &= \pi_{ABC}(\theta|y_{obs}). \end{aligned}$$

- (b) [10 marks] Consider a simple nonlinear climate model for convection rate (x_t) and horizontal (y_t) and vertical (z_t) temperature variation at time $t = 1, \dots, n$ running over n days. For parameters $(\tau, \rho, \beta, \sigma) \in [0, \infty)^p$ with $p = 4$, the evolution equations are

$$\begin{aligned} x_{t+1} &= x_t + c\tau(y_t - x_t) + \epsilon_{1,t}, \\ y_{t+1} &= y_t + cx_t(\rho - z_t) - cy_t + \epsilon_{2,t}, \\ z_{t+1} &= z_t + cx_t y_t - c\beta z_t + \epsilon_{3,t}, \end{aligned}$$

for $t = 1, \dots, n-1$, with c a small fixed positive constant which is known and $\epsilon_{i,t} \sim N(0, \sigma^2)$ independent and identically distributed for $i = 1, 2, 3$ and $t = 1, \dots, n-1$. Data $D_{obs} = (x_{obs}, y_{obs}, z_{obs})$ are gathered for a realisation of the process with $x_{obs} = (x_{obs,1}, \dots, x_{obs,n})$ and similarly for y_{obs} and z_{obs} . These data are a direct realisation of the evolution equations above. It is of interest to estimate the parameters τ, ρ, β and σ from the data D_{obs} .

- (i) The data space is $\mathcal{D} = \mathbb{R}^{3n}$. Give a summary statistic $S : \mathcal{D} \rightarrow \mathbb{R}^p$ which might be used in an ABC algorithm aiming to approximate the posterior $\pi(\tau, \rho, \beta, \sigma|D_{obs})$.

Solution: (S/4) Looking at the evolution equation for x_t we see we can estimate τ by fitting the simple regression $(x_{t+1} - x_t) \sim c(y_t - x_t)$ (linear model in R formula

notation). This will give an estimate $\hat{\tau}(D)$ for the slope and a first estimate for the standard error $\hat{\sigma}_1(D)$. Similarly regress $(y_{t+1} - y_t(1-c) + cx_t z_t) \sim cx_t$ to get fitted effect $\hat{\rho}(D)$ and standard error $\hat{\sigma}_2(D)$ and regress $(z_{t+1} - z_t - cx_t y_t) \sim cz_t$ to get $\hat{\beta}(D)$ and standard error $\hat{\sigma}_3(D)$. If we take $\hat{\sigma}(D) = c(\hat{\sigma}_1(D) + \hat{\sigma}_2(D) + \hat{\sigma}_3(D))/3$ and $S(D) = (\hat{\tau}, \hat{\rho}, \hat{\beta}, \hat{\sigma})$ then each component of $S(D)$ informs one parameter separately (actually consistent for $n \rightarrow \infty$).

Remark: There may be other reasonable answers. Our choice is in some ways so strongly informative we might give up on ABC and simply employ regression. However, if n is small then some form of Bayesian inference may help, and ABC is much simpler to implement here than MCMC.

- (ii) Let $D = (x, y, z)$ be a realisation of the process using some fixed parameters τ, ρ, β and σ and starting from $x_1 = x_{obs,1}, y_1 = y_{obs,1}$ and $z_1 = z_{obs,1}$. Explain how you would choose a suitable distance measure $d(S(D), S(D_{obs}))$ for use in ABC.

Solution: (S/2) The variance of components of $s = S(D)$ which take large values (for eg measured in millimetres rather than kilometres) will be much greater than those taking typically small values, and consequently variation in $d(s, s_{obs}) = |s - s_{obs}|$ will be driven by a small subset of parameters varying at large scale. We can fix this by centring and scaling the components of $S(D)$ to unit variance. If $S(D)$ is the resulting scaled statistic then we can take $d(S(D), S(D_{obs})) = |S(D) - S(D_{obs})|$ to be Euclidian distance.

- (iii) Suppose a parameter prior $\pi(\tau, \rho, \beta, \sigma)$ on $[0, \infty)^p$ is given. You may assume this is easily simulated. Write down a rejection-ABC algorithm to sample $\pi_{ABC}(\tau, \rho, \beta, \sigma | D_{obs})$.

Solution: (S/4) Let a threshold $\delta > 0$ be given.

1. Simulate $(\tau, \rho, \beta, \sigma) \sim \pi(\tau, \rho, \beta, \sigma)$; Take $x_1 = x_{obs,1}, y_1 = y_{obs,1}$ and $z_1 = z_{obs,1}$ and $\epsilon_{i,t} \sim N(0, \sigma^2)$ iid for $i = 1, 2, 3$ and $t = 1, \dots, n-1$.
2. Evaluate the evolution

$$\begin{aligned}x_{t+1} &= x_t + c\tau(y_t - x_t) + \epsilon_{1,t} \\y_{t+1} &= y_t + cx_t(\rho - z_t) - cy_t + \epsilon_{2,t} \\z_{t+1} &= z_t + cx_t y_t - c\beta z_t + \epsilon_{3,t}\end{aligned}$$

for $t = 1, \dots, n-1$ and set $D = (x, y, z)$.

3. Evaluate $d(S(D), S(D_{obs}))$ using the statistic S and distance d defined above.
4. If $d(S(D), S(D_{obs})) < \delta$ stop and return $(\tau, \rho, \beta, \sigma)$ else goto step 1.

Alternatively (and more practically) fix the number of simulations K in advance, save all the simulated parameter vectors $(\tau_i, \rho_i, \beta_i, \sigma_i)$ and corresponding $S(D_i)$, $i = 1, \dots, K$ values and use them to scale S to unit variance. Then compute all the $d_i = d(S(D_i), S(D_{obs}))$ values and take as samples the parameters $(\tau_i, \rho_i, \beta_i, \sigma_i)$ such that $d_i < d_{(\alpha K)}$, where $d_{(1)}, \dots, d_{(K)}$ are the order statistics of the d_i -values and α is the distance percentile we wish to target. This gives the

parameter values with the $100\alpha\%$ -smallest d -values and retains all the nice independence properties of rejection sampling. It is the same as choosing $\delta = d_{(\alpha K)}$ in rejection but is more automatic and adaptive.

- (c) [9 marks] We now return to the generic setting set out at the start of this question. Let $\mu(y) = E_{\theta \sim \pi(\cdot|y)}(\theta)$ and let $|s|$ denote Euclidean norm in $\mathbb{R}^p$.
- (i) Suppose we are doing ABC using $S(y) = \mu(y)$ and $d(s, s_{obs}) = |s - s_{obs}|$. Show that

$$|E_{\theta \sim \pi_{ABC}(\cdot|y_{obs})}(\theta) - \mu(y_{obs})| < \delta.$$

Solution: (N/4)

$$\begin{aligned} |E_{\theta \sim \pi_{ABC}(\cdot|y_{obs})}(\theta) - \mu(y_{obs})| &= \left| \int_{\mathbb{R}^p} \theta \pi_{ABC}(\theta|y_{obs}) d\theta - \mu(y_{obs}) \right| \\ &= \left| \int_{\mathbb{R}^p} \theta \left[\int_{\Delta_\delta(y_{obs})} \pi(\theta|y) p(y|Y \in \Delta_\delta(y_{obs})) dy \right] d\theta - \mu(y_{obs}) \right| \\ &= \left| \int_{\Delta_\delta(y_{obs})} \left[\int_{\mathbb{R}^p} \theta \pi(\theta|y) d\theta \right] p(y|Y \in \Delta_\delta(y_{obs})) dy - \mu(y_{obs}) \right| \\ &= \left| \int_{\Delta_\delta(y_{obs})} (\mu(y) - \mu(y_{obs})) p(y|Y \in \Delta_\delta(y_{obs})) dy \right| \\ &< \int_{\Delta_\delta(y_{obs})} |\mu(y) - \mu(y_{obs})| p(y|Y \in \Delta_\delta(y_{obs})) dy \\ &< \delta, \end{aligned}$$

since $\Delta_\delta(y_{obs}) = \{y \in \mathbb{R}^n : |\mu(y) - \mu(y_{obs})| < \delta\}$.

Remark: this idea (using $S(y) = \mu(y)$ gives $E_{\theta \sim \pi_{ABC}(\cdot|y_{obs})}(\theta) \rightarrow \mu(y_{obs})$ as $\delta \rightarrow 0$) is due to Fearnhead, P. and Prangle, D. (2012), "Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation". Journal of the Royal Statistical Society: Series B, 74: 419-474.

- (ii) Let $f(y; w)$ be a function with fixed parameters w which takes input y and returns as output an estimate of $\mu(y)$. Let $\hat{w}$ be an estimate for w chosen so that $f(y; \hat{w}) \simeq \mu(y)$. Suppose that for $t = 1, \dots, T$ we have samples $\theta_t \sim \pi(\cdot)$ and $y_t \sim p(\cdot|\theta_t)$ which are independent across pairs. Explain, and briefly justify, how you would use these samples to estimate w .

Solution: (N/3) Choose w to minimise

$$\widehat{\rho(\pi, w)} = \frac{1}{T} \sum_{t=1}^T (f(y_t; w) - \theta_t)^2$$

since $\widehat{\rho(\pi, w)}$ converges to the Bayes Risk $\rho(\pi, w)$ for the loss $(f(y; w) - \theta)^2$ and we know that is minimised by taking f to equal the posterior mean.

Remark: this idea (using a neural net to learn $\mu(y)$ and then using $S(y) = f(y; \hat{w})$ in ABC) is due to Bai Jiang, Tung-Yu Wu, Charles Zheng, Wing H. Wong

(2017), “Learning summary statistic for approximate Bayesian computation via deep neural network”, *Statistica Sinica*, Vol. 27: 1595-1618.

- (iii) Assemble these two results to give a generic rejection-ABC algorithm which learns its own summary statistics and uses them to do ABC.

Solution: (S/2) Fix $T > 0$ large enough for a reasonably precise estimate of w .

1. For $t = 1, \dots, T$ sample $\theta_t \sim \pi(\cdot)$ and $y_t \sim p(\cdot|\theta_t)$ jointly iid.
2. Set $\hat{w} = \arg \min_w \frac{1}{T} \sum_{t=1}^T (f(y_t; w) - \theta_t)^2$
3. Take $S(y) = f(y; \hat{w})$. Do ABC, ie, fix $\delta > 0$ and use rejection with $d(S(y), S(y_{obs})) = |S(y) - S(y_{obs})|$.
 - (a) Simulate $\theta \sim \pi(\cdot)$ and $y \sim p(\cdot|\theta)$
 - (b) If $y \in \Delta_\delta(y_{obs})$ stop and return θ else goto step 3(a).

The same remarks made in the solution to Q2(b)iii apply here.

2. Let $[L, U]$ be a finite interval and $\lambda : [L, U] \rightarrow \mathbb{R}^+$. In an inhomogeneous Poisson point process the probability for an event in a small interval δt is $\lambda(t)\delta t + o(\delta t)$ independently over intervals. If $Y = (Y_1, \dots, Y_N)$ are the event times in $[L, U]$ then the sample space for Y is $\Omega = \cup_{n=0}^{\infty} \Omega_n$ where $\Omega_0 = \{\emptyset\}$ and

$$\Omega_n = \{y \in [L, U]^n : L < y_1 < y_2 < \dots < y_n < U\}.$$

- (a) [10 marks] The probability density to realise $Y = y$, $y = (y_1, \dots, y_n)$ is

$$p(y|\lambda) = \exp(-\Lambda) \prod_{i=1}^n \lambda(y_i),$$

where $\Lambda = \int_L^U \lambda(t) dt$.

- (i) Let $Y \sim p(\cdot|\lambda)$ with $Y = (Y_1, \dots, Y_N)$. Show that $N \sim \text{Poisson}(\Lambda)$.

Solution: (B/4) $\Pr(N = n) = \Pr(Y \in \Omega_n)$ so

$$\begin{aligned} \Pr(N = n) &= \int_{\Omega_n} p(y|\lambda) dy \\ &= \exp(-\Lambda) \int_L^U \lambda(y_n) \int_L^{y_n} \lambda(y_{n-1}) \dots \int_L^{y_2} \lambda(y_1) dy_1 dy_2 \dots dy_n \\ &= \exp(-\Lambda) \int_L^U \lambda(t_n) \int_L^U \lambda(t_{n-1}) \dots \int_L^U \lambda(t_1) dt_1 dt_2 \dots dt_n \times \frac{1}{n!} \\ &= \exp(-\Lambda) \Lambda^n / n! \end{aligned}$$

The integral over the unconstrained $t \in [L, U]^n$ can be written as a sum of integrals over the spaces constrained to respect one of the $n!$ different orders. All these integrals are equal so when we remove the constraint we just divide by $n!$.

- (ii) Show that $p(y|\lambda)$ is normalised over Ω , that is $\Pr(Y \in \Omega|\lambda) = 1$.

Solution: (B/2)

$$\begin{aligned} \Pr(Y \in \Omega|\lambda) &= \sum_{n=0}^{\infty} \int_{\Omega_n} p(y|\lambda) dy \\ &= \sum_{n=0}^{\infty} \exp(-\Lambda) \Lambda^n / n! \\ &= 1 \end{aligned}$$

as the Poisson distribution is normalised.

- (iii) Suppose $N = n$ is fixed. Consider an MCMC algorithm targeting $p(y|\lambda, n)$ at fixed $n > 1$. Let $y_0 = L$ and $y_{n+1} = U$. The proposal selects a component $i \sim U\{1, \dots, n\}$ uniformly, samples $y'_i \sim U(y_{i-1}, y_{i+1})$ and proposes $y' = (y_1, \dots, y_{i-1}, y'_i, y_{i+1}, \dots, y_n)$. Calculate the acceptance probability.

Solution: (S/4) The proposal is symmetric $q(y'|y) = q(y|y')$ and the conditional

is proportional to the joint $p(y|\lambda, n) \propto p(y|\lambda)$ so the AP is

$$\begin{aligned}\alpha(y'|y) &= \min \left\{ 1, \frac{p(y'|\lambda, n)}{p(y|\lambda, n)} \right\} \\ &= \min \left\{ 1, \frac{\exp(-\Lambda) \prod_{j=1}^n \lambda(y'_j)}{\exp(-\Lambda) \prod_{j=1}^n \lambda(y_j)} \right\} \\ &= \min \left\{ 1, \frac{\lambda(y'_i)}{\lambda(y_i)} \right\}\end{aligned}$$

- (b) [5 marks] For $j = 1, \dots, J$ let $L < b_j < U$ be the birth year of individual j and let $\lambda(t)$ be the rate at which individuals catch a certain disease at time $L < t < U$. In a small interval δt an individual catches the disease with probability $\lambda(t)\delta t + o(\delta t)$ independently over time and people.

- (i) Calculate the probability $p_j(\lambda)$ that individual j catches the disease at least once over the interval $[b_j, U]$.

Solution: (S/3) Let N_j be the number of times j catches the disease. They were alive in the period $[b_j, U]$ so $N_j \sim \text{Poisson}(\Lambda_j)$ with

$$\Lambda_j = \int_{b_j}^U \lambda(t) dt$$

We want the probability for $N_j > 0$. This is $p_j(\lambda) = 1 - \exp(-\Lambda_j)$.

- (ii) A survey at time U tests J individuals alive at time U to see if they had caught the disease by that time. The survey records $z_j = 0$ if they never had the disease and $z_j = 1$ if they had it at least once. Let $z = (z_1, \dots, z_J)$. Show that

$$p(z|\lambda) = \prod_{j=1}^J p_j(\lambda)^{z_j} (1 - p_j(\lambda))^{1-z_j}.$$

Solution: (S/2) $z_j \sim \text{Bernoulli}(p_j)$ independently for each $j = 1, \dots, J$ giving the expression for $p(z|\lambda)$.

- (c) [10 marks] Continuing the problem stated in Question 2(b), suppose for $j = 1, \dots, J$ we have data $z_j \in \{0, 1\}$ and birth date b_j . We wish to infer $\lambda(t)$ for $L < t < U$. Let $\phi(t; a, b^2)$ be a normal density with mean a and variance b^2 . Consider a normal mixture

$$\lambda(t) = \sum_{k=1}^K w_k \phi(t; \mu_k, \sigma_k^2),$$

with K components with means $L < \mu_k < U$, standard deviations $\sigma_k > 0$ and component weights $w_k > 0$, $k = 1, \dots, K$. Let $\mu = (\mu_1, \dots, \mu_K)$, $\sigma = (\sigma_1, \dots, \sigma_K)$ and $w = (w_1, \dots, w_K)$. Assume independent priors $\pi_\mu(\mu)$, $\pi_\sigma(\sigma)$ and $\pi_w(w)$ are given with $\pi_\theta(\theta) = \prod_k \pi_\theta(\theta_k)$ when θ is μ, σ or w . The prior for the number of components is $\pi_K(k) = \text{Poisson}(k; m)$ with $m > 0$ is a fixed prior hyperparameter.

- (i) Write down the posterior $\pi(\mu, \sigma, w|z)$ (with $(b_1, \dots, b_J)$ known and fixed) and give the space on which it is defined.

Solution: (N/5) Since K is Poisson we could in principle have $K = 0$ components in which case $\lambda(t) = 0$ for all t . Then $p_j(\lambda) = 0$ (as $\Lambda_j = 0$ in $p_j(\lambda)$) so the likelihood will be zero unless all the z 's are zero. However, it is still well-defined and a possible outcome of the experiment so we should include it. If $\theta = (\mu, \sigma, w)$ then the space for θ when $K = k$ is

$$\Theta_k = [L, U]^k \times [0, \infty)^k \times [0, \infty)^k$$

with $\Theta_0 = \{\emptyset\}$ and the full space for θ is $\Theta = \cup_{k=0}^{\infty} \Theta_k$.

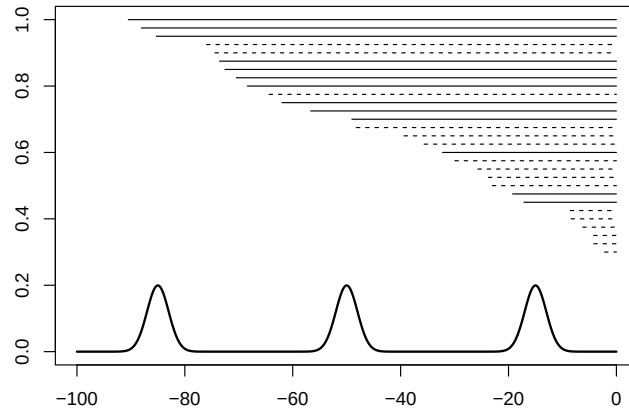
Here μ etc are (unordered) sets not vectors (ie ordered sets) so for any vectors $(\mu, \sigma, w) \in \Omega_k$ we have to multiply by $k!$ as any ordering is the same set. We could alternatively take a fixed order (for example, $\mu \in M_k = \{\mu \in [L, U]^k : \mu_1 < \mu_2 < \dots < \mu_k\}$) but then $\pi(\mu, \sigma, w | \mu \in M_k) = \pi(\mu, \sigma, w) / \pi(\mu \in M_k)$ with $\pi(\mu \in M_k) = 1/k!$ and we get the same thing.

Let $\lambda(\mu, \sigma, w)$ be given at each t by the formula in the question and let $p_j(\lambda(\mu, \sigma, w))$ be computed from this λ -function. The posterior is

$$\begin{aligned} \pi(\theta|z) &\propto p(z|\lambda(\mu, \sigma, w))\pi(\mu, \sigma, w)\pi(k)k! \\ &= \prod_{j=1}^J p_j(\lambda(\mu, \sigma, w))^{z_j} (1 - p_j(\lambda(\mu, \sigma, w)))^{1-z_j} \pi_{\mu}(\mu) \pi_{\sigma}(\sigma) \pi_w(w) m^k \end{aligned}$$

up to constants not depending on θ or k .

Remark: The setup for this question is illustrated in the figure.



Example with $L = -100$, $U = 0$. x -axis is time, y -axis is λ . The function $\lambda(t)$ is plotted in thick lines along the bottom. In this example there were three outbreaks of the disease. For each person $j = 1, \dots, J$ there is one horizontal bar running from b_j to U (from their birth to the present). The bar is solid if they caught the disease at least once and otherwise it is dashed. This approach was explained to me by Dr Ben Lambert. It is interesting that it is even possible to infer the history of epidemic outbreaks from age and sero-positivity data.

- (ii) Give proposals and acceptance probabilities for a reversible jump MCMC algorithm targeting $\pi(\mu, \sigma, w|z)$ (just give updates adding and deleting single components).

Solution: (N/5) Suppose the MC state is $X_t = (\mu, \sigma, w)$ and let k give the number of components in each vector. With probability one half propose to add and otherwise delete.

Add propose $\mu'_{k+1} \sim \pi_\mu(\cdot)$, $\sigma'_{k+1} \sim \pi_\sigma(\cdot)$ and $w'_{k+1} \sim \pi_w(\cdot)$. Set

$$\mu' = (\mu_1, \dots, \mu_k, \mu'_{k+1})$$

and similarly for σ' and w' . The acceptance probability for $\theta' = (\mu', \sigma', w')$ given $\theta = (\mu, \sigma, w)$ is

$$\begin{aligned} \alpha(\theta'|\theta) &= \min \left\{ 1, \frac{\pi(\mu', \sigma', w'|z)(k+1)^{-1}}{\pi(\mu, \sigma, w|z)\pi_\mu(\mu'_{k+1})\pi_\sigma(\sigma'_{k+1})\pi_w(w'_{k+1})} \right\} \\ &= \min \left\{ 1, \frac{p(z|\lambda(\mu', \sigma', w'))}{p(z|\lambda(\mu, \sigma, w))} \times \frac{m}{k+1} \right\}. \end{aligned}$$

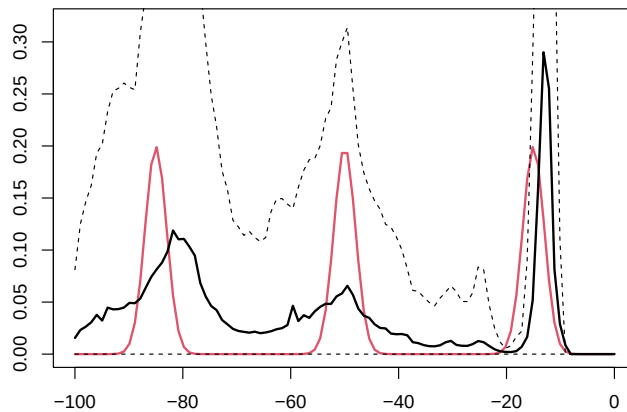
Other proposals are possible but this is tidy. Whatever is used the AP must match.

delete If $k = 0$ set $\alpha = 0$ and move to the accept/reject step. Otherwise select a component to delete $i \sim U\{1, \dots, k\}$. Set $\mu' = \mu_{-i}$, $\sigma' = \sigma_{-i}$ and $w' = w_{-i}$. The acceptance probability for $\theta' = (\mu', \sigma', w')$ given $\theta = (\mu, \sigma, w)$ is

$$\begin{aligned} \alpha(\theta'|\theta) &= \min \left\{ 1, \frac{\pi(\mu', \sigma', w'|z)\pi_\mu(\mu_i)\pi_\sigma(\sigma_i)\pi_w(w_i)}{\pi(\mu, \sigma, w|z)k^{-1}} \right\} \\ &= \min \left\{ 1, \frac{p(z|\lambda(\mu', \sigma', w'))}{p(z|\lambda(\mu, \sigma, w))} \times \frac{k}{m} \right\}. \end{aligned}$$

Now with probability $\alpha(\theta'|\theta)$ set $X_{t+1} = \theta'$ and otherwise $X_{t+1} = \theta$.

Remark: I implemented this to check it. I checked I got back the priors for μ, σ, w and k when $p(z|\lambda(\mu, \sigma, w))$ is set to equal one. Running it with $J = 100$ z -samples simulated using the true λ in the previous figure it did OK.



The solid black line is the posterior mean $\hat{\lambda}(t) = A^{-1} \sum_{a=1}^A \lambda(\mu^{(a)}, \sigma^{(a)}, w^{(a)})$ where $\mu^{(a)}, \sigma^{(a)}, w^{(a)}$, $a = 1, \dots, A$ are my posterior samples. The dashed lines are a 95% HPD set and otherwise as the figure above. We see it has correctly found the three “epidemics” although the signal is a bit weak.

3. (a) [12 marks] Let $\alpha > 0$ be given and let $H(dx) = h(x)dx$ be a continuous distribution with density $h(x)$ for $x \in \mathbb{R}^p$. Let $\theta = (\theta_1, \dots, \theta_n)$, $\theta_i \in \mathbb{R}^p$. Let $G \sim \Pi(\alpha, H)$ be a realisation of a Dirichlet Process (DP) and let $\theta_i | G, \alpha, H \sim G$ independently for $i = 1, \dots, n$.
- (i) Write down the definition of the DP in terms of the Dirichlet distribution.

Solution: (B/2) $G \sim \Pi(\alpha, H)$ is a DP iff for all partitions $A_1, \dots, A_r$ of $\mathbb{R}^p$ (such that $H(A_k)$, $k = 1, \dots, r$ exists), we have

$$G(A_1), \dots, G(A_r) \sim \text{Dirichlet}(\alpha H(A_1), \dots, \alpha H(A_r)).$$

In the course the space wasn't specifically $\mathbb{R}^p$ but that is fine given the setup in the question.

- (ii) Let $A \subseteq \mathbb{R}^p$ and $H(A) = \int_A h(x)dx$. Show that if $G \sim \Pi(\alpha, H)$ then $E(G(A)) = H(A)$.
Hint: if $X \sim \text{Beta}(u, v)$ then $E(X) = u/(u + v)$.

Solution: (B/2) By the definition of the DP,

$$G(A) \sim \text{Beta}(\alpha H(A), \alpha(1 - H(A))),$$

$$\text{so } E(G(A)) = \alpha H(A) / (\alpha H(A) + \alpha(1 - H(A))) = H(A).$$

- (iii) In the Blackwell-MacQueen urn process realising $\theta = (\theta_1, \dots, \theta_n)$,

$$\theta_i | \theta_{<i}, \alpha, H \sim \sum_{j=1}^{i-1} \frac{1}{\alpha + i - 1} \delta_{\theta_j} + \frac{\alpha}{\alpha + i - 1} H,$$

for $i = 1, \dots, n$, so $\theta_i \sim H$ is a random draw from H with probability $\alpha/(\alpha + i - 1)$ and otherwise θ_i is chosen randomly from $\theta_{<i} = (\theta_1, \dots, \theta_{i-1})$. Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ give the unique values in θ and let $S = (S_1, \dots, S_K)$ be a partition of $[n] = \{1, 2, \dots, n\}$ with $S_k = \{i \in [n] : \theta_i = \theta_k^*\}$ for $k \in [K]$.

Let $\pi(\theta^*, S) = \pi(\theta^* | S) P_{\alpha, [n]}(S)$. Show that $\pi(\theta^* | S) = \prod_{k=1}^K h(\theta_k^*)$ and

$$P_{\alpha, [n]}(S) = \alpha^K \prod_{k=1}^K (n_k - 1)! \prod_{i=1}^n \frac{1}{\alpha + i - 1},$$

where $n_k = |S_k|$ is the number of elements in S_k , $k \in [K]$.

Hint: the Blackwell-MacQueen process is equivalently a Chinese Restaurant Process.

Solution: (B,5) Write down the Blackwell-MacQueen urn process as a CRP.

1. Set $K = 1$, $S_1 = \{1\}$ and $S = \{S_1\}$. Simulate $\theta_1^* \sim H$ and set $\theta_1 = \theta_1^*$.
2. for $i = 1, \dots, n - 1$
with probability $\alpha/(\alpha + i)$ do (a) and otherwise do (b):
 - (a) Set $S_{K+1} = \{i + 1\}$, $S \leftarrow S \cup S_{K+1}$ and $K \leftarrow K + 1$. Simulate a new $\theta_K^* \sim H$ and set $\theta_{i+1} = \theta_K^*$.
 - (b) Let $n_k = |S_k|$ give the number of elements in S_k , $k = 1, \dots, K$; simulate $k \sim (n_1, \dots, n_K)/i$, set $S_k \leftarrow S_k \cup \{i + 1\}$ and $\theta_{i+1} = \theta_k^*$.

Each arrival $i = 1, \dots, n-1$ brings a denominator factor $(\alpha + i)^{-1}$, so the denominator is $\prod_{i=1}^{n-1} (\alpha + i) = \prod_{i=2}^n (\alpha + i - 1)$.

Now look at the numerator. Suppose at the end $S_k = \{i_1, i_2, \dots, i_{n_k}\}$. When i_1 arrived, i was equal $i_1 - 1$ and $k - 1$ tables were occupied, so i_1 creates a new set S_k via option 2a with probability $\alpha/(\alpha + i_1 - 1)$ (and we already accounted for the denominator).

After that, for $j = 2, \dots, n_k$, when i_j arrived $i = i_j - 1$ and set S_k had $j - 1$ elements, so i_j chose option 2b wp $(i_j - 1)/(\alpha + i_j - 1)$ and then set S_k wp $(j - 1)/(i_j - 1)$, so the factor is $(j - 1)/(\alpha + i_j - 1)$. Including the first arrival in this set, the overall numerator factor from table k is $\alpha(n_k - 1)!$.

If we end up with K tables then there are $K - 1$ events in which a new table is chosen so

$$\begin{aligned} P_{\alpha, [n]}(S) &= \alpha^{K-1} \prod_{k=1}^K (n_k - 1)! \prod_{i=2}^n \frac{1}{\alpha + i - 1} \\ &= \alpha^K \prod_{k=1}^K (n_k - 1)! \prod_{i=1}^n \frac{1}{\alpha + i - 1}. \end{aligned}$$

All the θ_k^* , $k \in [K]$ are independent draws from H is given K (ie, given S) we have $\pi(\theta^*|S) = \prod_{k=1}^K h(\theta_k^*)$.

- (iv) Let $n=100$. Calculate the prior probability $\theta_1 = \theta_{50} = \theta_{100}$.

Solution: (S,3) For equal θ 's we need 1, 50 and 100 to be in the same partition set, by the continuity of H . Since $1 \in S_1$, this is the probability $50, 100 \in S_1$. Arrivals are exchangeable so this is the same as the probability 1, 2 and 3 are in the same set, which is $1/(\alpha + 1) \times 2/(\alpha + 2)$ from the CRP.

- (b) [13 marks] Suppose the data $y_b = (y_{b,1}, \dots, y_{b,n_b})$, $b = 1, \dots, B$ come in B blocks. Consider a hierarchical Dirichlet process model for these data. Let α and H be given as above and let $G_0 \sim \Pi(\alpha, H)$. Let

$$G_b|G_0 \sim \Pi(\beta, G_0), \quad b \in [B],$$

be B independent realisations of a DP with parameter $\beta > 0$ and base distribution G_0 . For $i \in [n_b]$ let $\theta_{b,i}|G_b \sim G_b$ be conditionally independent given G_b (and notice $\theta_{b,i}|G_b, G_0 \sim \theta_{b,i}|G_b$).

- (i) Show that marginally $\theta_{b,i}|G_0 \sim G_0$ for each $b \in [B]$ and $i \in [n_b]$ and hence show that the atoms of G_b are a subset of the atoms of G_0 .

Hint: Assume that G_b and G_0 are discrete distributions.

Solution: (S/2) for $\theta_{b,i}|G_0 \sim G_0$ and (N/2) for the subset relation. We have

$$\begin{aligned} \Pr(\theta_{b,i} \in A|G_0) &= E(\mathbb{I}_{\theta_{b,i} \in A}|G_0) \\ &= E(E(\mathbb{I}_{\theta_{b,i} \in A}|G_b)|G_0) \\ &= E(G_b(A)|G_0). \end{aligned}$$

However, if $G_b|G_0 \sim \Pi(\beta, G_0)$ then $E(G_b(A)|G_0) = G_0(A)$ by Q3(a)ii. Since

$\Pr(\theta_{b,i} \in A | G_0) = G_0(A)$ for all sets A such that $G_0(A)$ exists, we have (A) $\theta_{b,i} | G_0 \sim G_0$.

If $\theta_{b,i} \sim G_b$ with $G_b \sim \Pi(\alpha, G_0)$ then G_b must have an atom at $\theta_{b,i}$ as it is almost surely discrete. However we also have $\theta_{b,i} \sim G_0$ marginally by (A). Since G_0 is also discrete, $\theta_{b,i}$ must also be an atom of G_0 so any atom of G_b is an atom of G_0 .

- (ii) For $b \in [B]$ let $S_b = (S_{b,1}, \dots, S_{b,K_b})$ be a partition of $\{b\} \times [n_b]$ and let

$$\mathcal{T} = \bigcup_{b \in [B]} \{b\} \times [K_b],$$

be the set of all pairs (b, k) , $b \in [B]$ and $k \in [K_b]$. For each $(b, k) \in \mathcal{T}$ let $\theta_{b,k}^* \sim G_0$. Let $T = (T_1, \dots, T_L)$ be a partition of $\mathcal{T}$ and for $\ell \in [L]$ let $\theta_\ell^{**} \sim H$. If for some $(b, k) \in \mathcal{T}$ it holds that $(b, i) \in S_{b,k}$ and $(b, k) \in T_\ell$ then $\theta_{b,i} = \theta_{b,k}^* = \theta_\ell^{**}$. Calculate the prior $\pi(\theta^{**}, S, T)$ for the random partitions and parameters.

Solution: (N/4) This is basically the same as before but we add a layer to the hierarchy. In the CRP metaphor we now have B “restaurants”, one for each block, and when the process is finished we will have K_b tables occupied in restaurant b . The “dish” served at table $k \in [K_b]$ will be $\theta_{b,k}^*$ so each table has a parameter vector. We bring in the customers (b, i) one at a time and seat them at tables (b, k) , $k \in [K_b]$, so when we are done $S_b = (S_{b,1}, \dots, S_{b,K_b})$ will be a partition of all the elements $\{b\} \times [n_b]$ in block b and $\theta_{b,i} = \theta_{b,k}^*$ for $(b, i) \in S_{b,k}$.

The main new feature is that when a customer opens a new table we don’t simulate the new $\theta_{b,k}^*$ from H , we simulate it from G_0 . There is one (restaurant, table) vector $\theta_{b,k}^*$ for each $(b, k) \in \mathcal{T}$, and since they are draws from G_0 , the set $\mathcal{T}$ is partitioned into groups $T = (T_1, \dots, T_L)$. All the tables $(b, k) \in T_\ell$ have the same dish $\theta_\ell^{**} \in Q$, which is an independent draw from H , so we set $\theta_{b,k}^* = \theta_\ell^{**}$ for all $(b, k) \in T_\ell$. It follows that

$$\pi(\theta^{**}, S, T | y) = P_{\alpha, \mathcal{T}}(T) \prod_{b=1}^B P_{\beta, \{b\} \times [n_b]}(S_b) \prod_{\ell=1}^L h(\theta_\ell^{**}).$$

It would be clearer to write the hierarchical Blackwell-MacQueen urn process as an algorithm but we don’t really need that as we know the partitions are realised by CRPs $P_{\alpha, \mathcal{T}}(T)$ and $P_{\beta, \{b\} \times [n_b]}(S_b)$ etc.

Remark: this is Teh, Y. W., Jordan, M. I., Beal, M. J., & Blei, D. M. (2006). “Hierarchical Dirichlet Processes”, *JASA*, 101(476), 1566–1581. The classical application, given in the paper, is to topic-models, so $y_b = (y_{b,1}, \dots, y_{b,n_b})$ is a book with n_b words and there are B books in the library. The vectors θ_ℓ^{**} are topics, each topic has a distribution $\theta_\ell^{**} = (\theta_{\ell,1}^{**}, \dots, \theta_{\ell,V}^{**})$ over the V possible words in the vocabulary (for example if the topic is “royalty” there is a high probability for the word “coronation” and a low probability for the word “badger” and the converse if the topic is British wildlife.). The vector $\theta_{b,i}$ is a probability vector for word i in book b so $y_{b,i} \sim \text{Cat}(\theta_{b,i})$ and G_b is the distribution over topics in book b and G_0 is the distribution over all topics in all books in the library.

- (iii) Suppose $B = 2$ and $n_1 = n_2 = 100$. What is the probability for $\theta_{1,1} = \theta_{1,50} = \theta_{2,100}$?

Solution: (N/3) Sort the arrivals so $(1, 1), (1, 50)$ and $(2, 100)$ are assigned θ -values first, in that order, using the exchangeability of both the T and S processes. We know $(1, 1) \in S_{1,1}$ and $(2, 100) \in S_{2,1}$ so $\theta_{b=1,i=1} = \theta_{b=1,k=1}^*$ and $\theta_{b=2,i=100} = \theta_{b=2,k=1}^*$. We also know $(b = 1, k = 1) \in T_1$ and so $\theta_{1,1}^* = \theta_1^{**}$. Then

$$\begin{aligned}
\Pr(\theta_{1,1} = \theta_{1,50} = \theta_{2,100}) &= \Pr(\theta_{1,1} = \theta_{1,50} = \theta_{2,100} | (1, 50) \in S_{1,1}) \Pr((1, 50) \in S_{1,1}) \\
&\quad + \Pr(\theta_{1,1} = \theta_{1,50} = \theta_{2,100} | (1, 50) \in S_{1,2}) \Pr((1, 50) \in S_{1,2}) \\
&= \Pr(\theta_{1,1}^* = \theta_{2,1}^*) \frac{1}{\beta + 1} + \Pr(\theta_{1,1}^* = \theta_{1,2}^* = \theta_{2,1}^*) \frac{\beta}{\beta + 1} \\
&= \Pr((b = 2, k = 1) \in T_1) \frac{1}{\beta + 1} \\
&\quad + \Pr((b = 1, k = 2) \in T_1, (b = 2, k = 1) \in T_1) \frac{\beta}{\beta + 1} \\
&= \frac{1}{\alpha + 1} \frac{1}{\beta + 1} + \frac{1}{\alpha + 1} \frac{2}{\alpha + 2} \frac{\beta}{\beta + 1}
\end{aligned}$$

- (iv) The observation model is $y_{b,i} \sim p(\cdot | \theta_{b,i})$ jointly independent for $b \in [B]$ and $i \in [n_b]$. Give the posterior $\pi(\theta^{**}, S, T | y)$ up to an overall normalising constant.

Solution: (N/2) In the observation model for $y_{b,i}$ when $(b, i) \in S_{b,k}$ and $(b, k) \in T_\ell$ we have $\theta_{b,i} = \theta_\ell^{**}$ so $y_{b,i} \sim p(\cdot | \theta_\ell^{**})$ and

$$\pi(\theta^{**}, S, T | y) = P_{\alpha, \tau}(T) \prod_{b=1}^B P_{\beta, \{b\} \times [n_b]}(S_b) \prod_{\ell=1}^L h(\theta_\ell^{**}) \prod_{(b,k) \in T_\ell} \prod_{(b,i) \in S_{b,k}} P(y_{b,i} | \theta_\ell^{**}).$$